

**Manipulating Historical Narratives, Implicit Association Test and Ingroup
Bias: in Japan-Korea Colonial History Case**

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Abstract

The Implicit Association Test (IAT) has been widely applied to measure implicit biases across various dimensions, including race, gender, age, and sexual orientation. This study demonstrates the effectiveness of IAT in distinguishing attitudinal preferences between Japanese and Korean individuals. Furthermore, as a third cultural group, Taiwanese college students showed no significant implicit attitudinal differences towards these two nationalities. Therefore, this study explores how interpretations of colonial history might manipulate emotional inclinations and group identification, consequently influencing subconscious attitudinal responses. Using experimental manipulation of video content and the Implicit Association Test (IAT), this study examines how historical narrative films influence implicit attitudes among Taiwanese participants.

Keywords: *Implicit Association Test (IAT), historical narratives, Japan-Korea Colonial History, Nationalities preference*

Introduction

Anthony G. Greenwald and Mahzarin R. Banaji (1995) first proposed a new perspective in social cognition: "implicit social cognition." Implicit social cognition posits that people's attitudes, beliefs, and stereotypes often reside at an unconscious level and may not be revealed through self-report. Implicit social cognition connects with important concepts from the works of many early psychologists, such as the influence of psychoanalytic theories of the unconscious. Similar concepts were also introduced into the behaviorist school (Dollard & Miller, 1950; Doob, 1947; Osgood, 1957). These theories, which previously lacked empirical support, were later scientifically validated, with cognitive psychologists successfully verifying the effectiveness of unconscious cognitive experiments in various studies (Greenwald, 1992; Kihlstrom, 1987).

Subsequently, Greenwald et al. (1998) developed the "Implicit Association Test" (IAT) to measure the strength of automatic associations between specific concepts and evaluations (positive/negative). By measuring differences in participants' reaction times during categorization tasks, the IAT can infer the underlying preferences or biases. In the congruent trials of the test, participants categorize stimuli in alignment with their existing implicit preferences, resulting in smaller reaction errors when classifying in the same direction (for instance, responding similarly to names of their own nationalities and positive evaluation words; responding similarly to names of other nationalities and negative evaluations). In contrast, in incongruent trials, participants categorize stimuli in a way that is inconsistent with their own attitudes (e.g., responding in the same direction to names of their own nationalities and negative evaluations, and therefore showing larger reaction time differences due to incompatibility with implicit preferences) (Kurdi et al., 2019).

This study focuses on ethnic groups as subjects of observation, specifically measuring attitudes towards Japanese and Korean ethnic groups. After initially measuring the attitudes of Taiwanese participants to determine any preferences towards Japan and Korea, researchers manipulated historical interpretations for Taiwanese participants to observe whether their implicit biases change as a result of the video manipulation. Both Korea and Taiwan were once under Japanese colonies rule and were influenced by Japanese infrastructure and policies. However, Taiwan does not appear to hold the same negative attitudes and sentiments towards Japan as Korea does. This discrepancy may be due to complex historical backgrounds, national characteristics, national consciousness, and various political factors. On the 50th anniversary of Korea's liberation, President Kim Young-sam ordered the demolition of the Government-General of Chōsen Building, also known as the Japanese General Government Building, under the pretext of "eradicating symbols of Japanese colonial rule," and most other Japanese colonial buildings in Korea were also demolished under the Korean government's directive (Chen, 2013). In the early 20th century, the Korean Peninsula was under Japanese colonial rule, and a large number

of Koreans were forcibly conscripted and mobilized for labor during World War II. This historical experience profoundly shaped Korean society's national consciousness and anti-Japanese sentiment, with Koreans referring to this period as the "Imperial Japanese compulsive occupation period," or "The Dark Japanese Imperial Period."

Method

Participants

Three distinct participant groups were recruited for this study:

Taiwanese Participants

Participants ($N = 32$) were undergraduate students enrolled in psychology courses at National Cheng Kung University, Taiwan. Recruitment followed two schemes:

- *Required participation:* Students in "Psychological Experiment (II)" fulfilled course requirements through study involvement
- *Voluntary participation:* Students from "General Psychology" and "Introduction to Psychology" courses could receive extra credit for participation.

International Participants

Japanese ($N = 8$) and Korean ($N = 4$) participants were recruited through:

- Contacting with international student associations at National Cheng Kung University
- Snowball sampling, where participants could refer acquaintances for participation.

There are monetary compensation of NT\$200 (approximately USD\$6) for international participants per completed session. All participants provided informed consent.

Implicit Association Test (IAT) design

The IAT procedure consists of five steps. First, participants are exposed to a target concept identification task, that requires them to classify stimuli into two categories (in this case, Korean surnames and Japanese surnames) using two different response keys. Secondly, they learned to categorize the evaluation stimuli, positive words, and negative ones. Third, the task combines the target concept and attribute categorization, requiring participants to alternately classify names and evaluative words. In the fourth step, the response keys for target categorization are swapped, meaning participants must use the opposite keys to categorize the target concepts. Finally, in the fifth step, the swapped target categorization is combined with the unchanged attribute categorization.

Participants in the study were divided into four groups based on nationality and manipulation received: Japanese and Korean participants, and Taiwanese participants subjected to pro-Japanese or pro-Korean manipulation. Within each group, participants were manipulated on one procedural variable: the initial response key assignment (left or right) for Korean and Japanese surnames. Each participant underwent two IAT measurements: one using truncated Japanese surnames as stimuli. The IAT procedure and stimulus design in this study followed the standard protocol proposed by Greenwald, McGhee, and Schwartz (1998).

Materials

Apparatus

The experiment was administered using PsychoPy v2024.2.4 (Peirce J. et al., 2019) running on either MacOS or WindowsOS laptop. Response collection utilized the laptop's built-in keyboard with the following mappings: Left response: 'A' key (left index finger), Right response: 'L' key (right index finger), and Trial advancement: Spacebar.

IAT Stimulus Materials

There are 100 different words stimuli in total for the categorization tasks, comprising the following four categories: 25 Korean surnames, 25 Japanese surnames, 25 positive connotation words and 25 negative connotation words.

Positive and negative words were selected based on the word standards established by Bellezza, Greenwald, and Banaji (1986). Korean and Japanese surnames were referenced from the standard IAT procedure designed by Greenwald, McGhee, and Schwartz (1998), with initial screening by two Korean and two Japanese judges to ensure their national representativeness and identifiability.

A complete list of words and surnames is provided in Appendix A.

Adjustment of Surname Length

Considering the differences in character count between Japanese and Korean surnames, this study, following Greenwald et al. (1998), truncated Japanese surnames to match the length of corresponding Korean surnames while retaining their linguistic features (e.g., "Kawabashi" was shortened to "Kawa" to match "Youn"). Participants first encountered full-length Japanese surnames, followed by the simplified versions. Therefore, each experiment included two IAT tests: IAT using full-length Japanese surnames and IAT using simplified Japanese surnames. To ensure the clear stimulus presentation, surnames were presented in uppercase letters and evaluation words were presented in lowercase letters.

Manipulation Materials

Taiwanese participants were divided into two groups: those exposed to a pro-Japanese stance manipulation and those exposed to a pro-Korean stance manipulation. After completing Experiment 1, each group received the corresponding manipulation, which consisted of AI-generated short videos lasting approximately five minutes. The URLs are provided in Appendix B.

Procedure

Trial Blocks

Each experimental block consisted of 50 trials. Before each block, participants viewed on-screen instructions for the upcoming categorization task (see Figure 1).

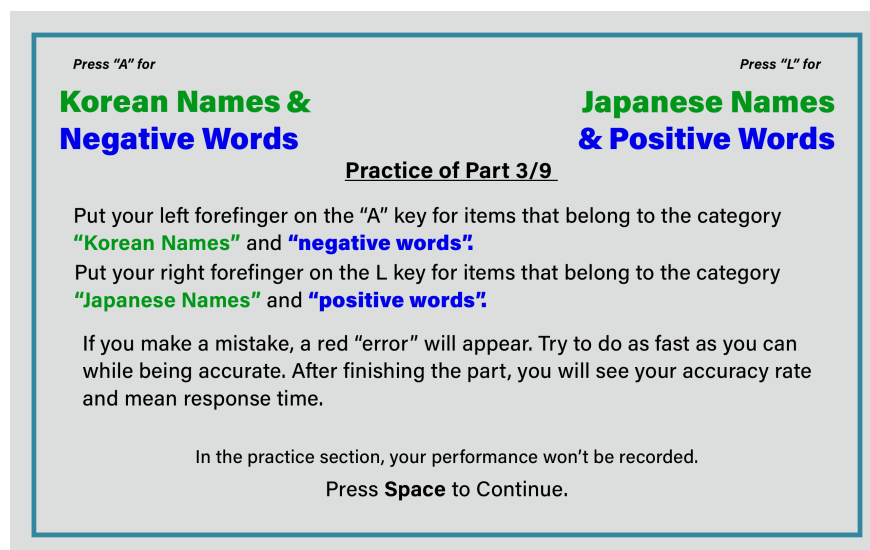


Figure 1: *Instructions for combined target and evaluation stimuli*

Whenever the task changed (for example, changing to a new categorization category), a 50-trial practice block was performed first, followed by data collection blocks. In the combined categorization task phase, two formal data collection blocks followed. Notably, For each participant, the surname categorization task only had a practice block during the initial configuration; subsequent transitions to surname categorization tasks did not include practice blocks.

Stimulus Presentation

The first trial in each block began 1.5 seconds after the category cues were displayed. Stimulus words were presented in black font on a light gray background, centered. Each stimulus remained on screen until the participant responded. After a response, the system delayed 250 milliseconds before presenting the next stimulus. If the participant made an error, the original stimulus was replaced by the word "error" for 300

milliseconds, and the inter-stimulus interval increased by 300 milliseconds. After completing each 50-trial block, feedback was provided, including the average reaction time (in milliseconds) and accuracy percentage for that block.

Stimulus Selection

Stimulus words were selected with evenly repetition times and randomization for each participant. In a single categorization task, each of the two categories contained 25 items, so each stimulus was used once in the 50 trials of each block. In the combined task, two blocks were recorded for each pairing method, totaling 100 trials. We constructed two experimental stimulus files (condition files) as follows: One file consisted of 25 targets, which divided as 2 part, 13 Japanese or Korean surnames and 12 opposing surnames; and 25 evaluative words, divided as 13 evaluative words paired with the aforementioned thirteen surnames and 12 opposing evaluative words. The second file consisted of the remaining words. Each of the two condition files were applied to one block for the combined task, ensuring that all 100 stimuli were evenly used across the two recorded blocks. Target concept stimuli and attribute stimuli appeared alternately.

Experiment Procedure in Phase 1

Each group of participants was divided into two subgroups based on initial key configuration, while each participant completed two IAT tests sequentially. For instance, in the first IAT, left key is assigned to Japanese surnames; Right key is assigned to Korean surnames for the first subgroup, and the other half underwent with opposite setting. Secondly, the first subgroup would underwent the same setting as the second subgroup's first IAT. See Figure 2 & 3 for the two kind of task sequences in each experiment.

Task	Content (Korean)	(L) Negative	(R) Positive
A	A-1 / A-2 / A-3	Japanese Surnames	Korean Surnames
B	B-1 / B-2	Negative words	Positive words
C	C-1 / C-2 / C-3	Japanese Surnames + Negative words	Korean Surnames + Positive words
D	D-1 / D-2	Korean Surnames	Japanese Surnames
E	E-1 / E-2 / E-3	Korean Surnames + Negative words	Japanese Surnames + Positive words
F	F-1 / F-2	Korean Surnames	Truncated Japanese Surnames
G	G-1 / G-2 / G-3	Korean Surnames + Negative words	Truncated + Positive words
H	H-1 / H-2	Truncated Japanese Surnames	Korean Surnames
I	I-1 / I-2 / I-3	Truncated + Negative words	Korean Surnames + Positive words

Figure 2: one of the trial blocks sequence type, including two IATs, 9 stages

Task	Content (Japan)	(L) Negative	(R) Positive
A	A-1 / A-2 / A-3	Korean Surnames	Japanese Surnames
B	B-1 / B-2	Negative words	Positive words
C	C-1 / C-2 / C-3	Korean Surnames + Negative words	Japanese Surnames + Positive words
D	D-1 / D-2	Japanese Surnames	Korean Surnames
E	E-1 / E-2 / E-3	Japanese Surnames + Negative words	Korean Surnames + Positive words
F	F-1 / F-2	Truncated Japanese Surnames	Korean Surnames
G	G-1 / G-2 / G-3	Truncated + Negative words	Korean Surnames + Positive words
H	H-1 / H-2	Korean Surnames	Truncated Japanese Surnames
I	I-1 / I-2 / I-3	Korean Surnames + Negative words	Truncated + Positive words

Figure 3: one of the trial blocks sequence type, including two IATs, 9 stages

The second IAT differed from the first one in three ways: First, the practice phase for evaluative words was omitted. Second, full-length Japanese surnames were replaced

with truncated versions. Last, the key configuration for target categorization was reversed compared to the first IAT.

There are 9 stages, 19 experimental trial blocks, in phase 1 experiment, which is marked by trial block codes, ranging from 1 to 19.

The first IAT comprised five stages: (a) Initial target categorization (Block 2); (b) Attribute categorization: Positive vs. Negative words (Block 4); (c) First combined task (Blocks 6 and 7); (d) Target categorization key reversal (Block 8); (e) Combined task after key reversal (Blocks 10 and 11).

The second IAT omitted the attribute categorization stage and consisted of four stages: (f) Initial target categorization (key configuration continued from (d), Block 12); (g) First combined task (Blocks 14 and 15); (h) Target categorization after key reversal (Block 16); (i) Combined task after key reversal (Blocks 18 and 19)

Experiment Procedure in Phase 2

Experiment in the second phase was conducted solely on Taiwanese participants. After completing Experiment in phase one and receiving the corresponding manipulation, the same complete experimental procedure as Experiment 1 was initiated for the second time after participants confirmed their ability to continue with the procedure.

Results

This study was divided into two experimental phases. Experiment in the first phase measured the implicit association of participants from different nationalities towards Japanese and Korean ethnic groups, as well as the initial implicit association of Taiwanese individuals before manipulation. Experiment in the second phase measured data from Taiwanese participants after historical narrative manipulation to observe changes in their implicit attitudes before and after the manipulation. The comparison of reaction time differences between congruent and incongruent tasks served as the core indicator for evaluating implicit preferences.

Experiment in phase 1

In the first phase, Japanese and Korean participants each exhibited clear in-group preference. Japanese participants responded more quickly in the categorization task pairing Japanese surnames with positive words, indicating a stronger automatic association between their own ethnic group and positive evaluations. Similarly, Korean participants showed similar performance, indicating a positive implicit attitude towards their own ethnic group. This ethnic consistency preference aligns with prior expectations regarding "in-group bias" in implicit social cognition (Greenwald et al., 1998). Tables 1 and 2 present detailed statistical data for the Japanese and Korean participant groups, respectively.

Table 1: One-Sample t-Tests for Korean Participants

IAT Type	Mean D	SD D	n	t	p
Full	0.504	0.149	4	6.770	0.007**
Truncated	0.464	0.217	4	4.284	0.023*

* $p < .05$, ** $p < .01$

Table 2: One-Sample t-Tests for Japanese Participants

IAT Type	Mean D	SD D	n	t	p
Full	0.450	0.387	8	3.289	0.013*
Truncated	0.371	0.343	8	3.061	0.018*

* $p < .05$, ** $p < .01$

Taiwanese participants in the same task did not show a significant preference as international participants did. Regardless of whether Japanese or Korean surnames were paired with positive words, the differences in participants' performance were not substantial, and changes in reaction time did not exhibit a clear direction. This result indicates that, in the absence of a profound historical stance, Taiwanese participants, acting as third-party observers, maintained a relatively neutral implicit association towards both the Japanese and Korean.

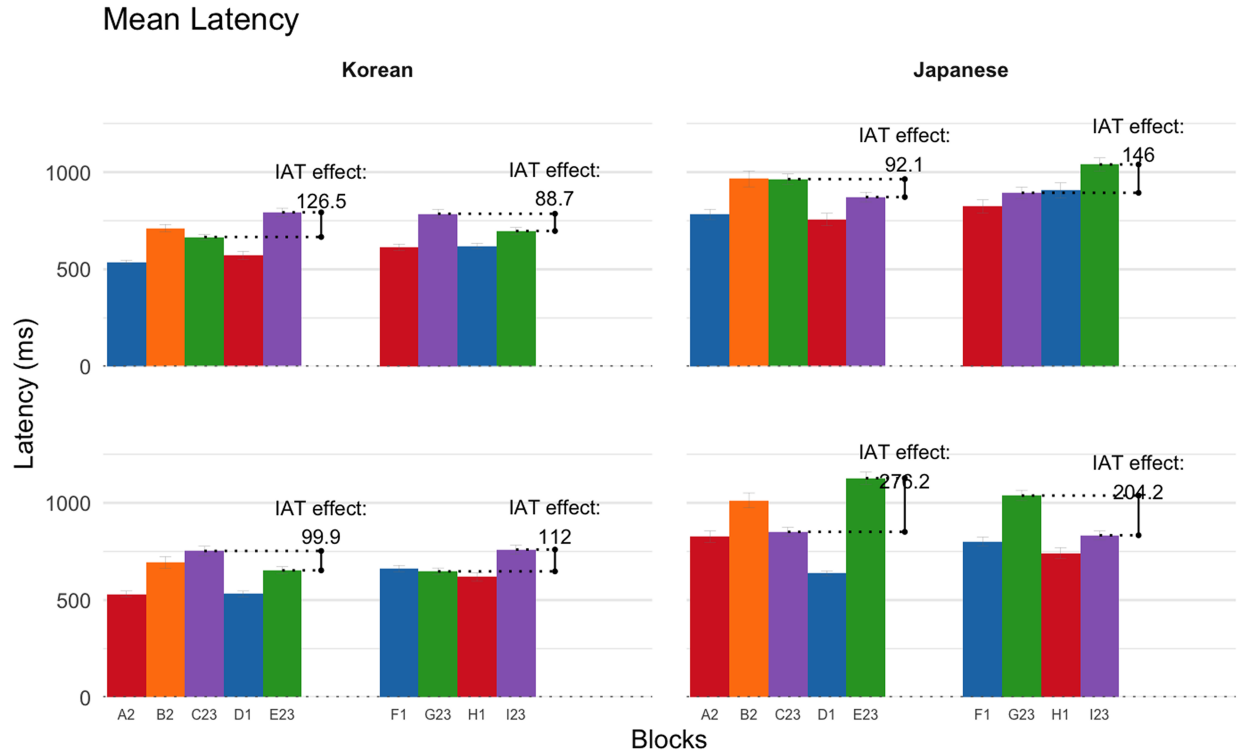


Figure 4: Mean (untransformed) latency results of Experiment in phase 1, separately for 4 Korean and 8 Japanese participants and for subjects who received the two orders of presentation. Error bars are within-cell standard deviations contributing to each mean.

Experiment in phase 2

After the first phase, Taiwanese participants were randomly assigned to watch films presenting two different historical perspectives. One group watched a historical narrative video with a pro-Japanese bias, while the other watched content with a pro-Korean stance. Both contents were AI-synthesized, describing different historical narratives and interpretations of the Japanese colonial period.

After manipulation, participants completed the same IAT test as in the first phase. However, There is no significant differences in the performance of the two groups in the second phase compared to the first. Regardless of the historical manipulation received, changes in reaction time and error rates under congruent and incongruent conditions did not show systematic alterations. Tables 3 and 4 present detailed statistical data for

Taiwanese participants after manipulation, showing that they maintained their original neutral attitude both before and after the manipulation, without exhibiting a clear direction of implicit preference.

Table 3: One-Sample t-Tests for pro-Korean Taiwanese Participants

IAT Type	Mean D	SD D	n	t	p
Full	0.099	0.425	16	0.932	0.366
Truncated	-0.003	0.396	16	-0.032	0.975

* $p < .05$, ** $p < .01$

Table 4: One-Sample t-Tests for pro-Japanese Taiwanese Participants

IAT Type	Mean D	SD D	n	t	p
Full	0.094	0.419	16	0.894	0.385
Truncated	0.051	0.333	16	0.612	0.549

* $p < .05$, ** $p < .01$

These results suggest that a single, short historical narrative manipulation did not cause a substantial adjustment effect on the implicit associations of Taiwanese participants.

Overall, experiment in phase 1 supports the clear existence of ethnic implicit bias in Japanese and Korean ethnic group, while Taiwanese participants tended towards neutrality under unmanipulated conditions. The results of Experiment 2, however, indicate that attempts to alter the implicit attitudes of third-party observers through short video manipulation may be limited by the stability of implicit associations and the lack of persuasiveness of the manipulation material.

Discussion

The results of Experiment in phase 1 indicate that Japanese and Korean participants exhibited significant differences in in-group preference, tending to hold more positive implicit attitudes towards their own ethnic groups. This aligns with Greenwald et

al.'s (1998) theory of in-group preference and supports the fundamental assumption in implicit social cognition regarding the connection between group belonging and latent attitudes. Conversely, Taiwanese participants in Experiment 1 did not display a clear preference for either the Japanese or Korean ethnic group, reflecting a cultural distance and historical emotional neutrality in their role as "third-party observers." This may be related to the relatively weak historical memory construction concerning Japan and Korea within the Taiwanese educational context (Bar-Tal & Salomon, 2006).

In the second phase of experiment, AI-generated historical narrative videos were used as manipulative stimuli to influence participants' emotional inclinations and implicit attitudes. However, participants' IAT responses did not show significant changes after viewing the videos. There might be three possible reasons for this unexpected outcome: the stability of implicit attitudes, the quality of the manipulative material, and the singularity of the short-term stimulus.

First, implicit attitudes possess high stability and resistance to manipulation. As Nosek et al. (2007) pointed out, implicit attitudes are often formed through long-term socialization processes and cannot be altered by a single stimulus.

Second, a possible reason for the ineffective manipulation is that the manipulative materials used in this study were all generated by artificial intelligence. Although they possessed informational completeness and visual persuasiveness, they lacked the emotional resonance common in human narratives. This might have prevented them from effectively triggering participants' emotional responses and deeper meaning construction, thus making it difficult to alter their subconscious cognitions.

Finally, the manipulation in this study involved a single source of information and a single video viewing, failing to present realistic multi-sensory or social interaction stimuli. Previous research has indicated that attitude change often requires repeated exposure, multi-channel input, and connections to real-life experiences to produce significant implicit attitude change effects (Petty & Cacioppo, 1986). Taiwanese participants' existing

knowledge and personal experiences regarding Japanese and Korean history might have formed a kind of "cognitive inertia," leading them to maintain their original implicit attitudes when faced with short-term stimuli.

However, given the widespread penetration of AI content into mass communication and educational environments, its effect on manipulating social attitudes still requires careful evaluation. The effect remains limited on psychological variables involving subconscious and emotional memory levels. The results of this study suggest that a single AI-produced historical narrative video is insufficient to cause significant cognitive change or emotional arousal for a group with established cognitive biases.

In summary, this study supports the view that implicit attitudes exhibit cultural differences and deep historical structures. It also indicates that AI-generated historical narratives, in the short term, do not significantly alter implicit attitudes. Future research could consider extending manipulation time, introducing more interactive scenarios, or combining in-depth interviews with qualitative analysis to more comprehensively explore the relationship among historical education, media narratives, and attitude change

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Appendix

A

The complete set of experimental materials described above has been made publicly available, including: The PsychoPy experimental program, Condition files for stimulus presentation, The full stimulus table containing all 100 words used in the categorization task.

These materials are archived in the open-access repository:

IAT PsychoPy Experiment Implementation:

https://github.com/yurenliuuni/IAT_psychopy_exp_ncku2025

B

Pro-Japanese stance manipulation film:

https://youtu.be/ly00bhWzef8?si=j3UeB_mJMLGgsXaD

Pro-Korean stance manipulation film:

<https://youtu.be/qpzzUx1t-3U?si=E1l7uKbWXcZNfefL>

C

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Conceptualization	V	V	V	V	V
Methodology	V	V	V		
Software	V	V	V		
Validation	V	V	V	V	V
Formal Analysis	V		V		
Investigation	V	V	V	V	V
Resources	V	V	V	V	V
Data Curation		V	V	V	V
Writing - Original Draft		V	V	V	V
Writing - Review & Editing	V				
Visualization	V	V	V	V	V
Supervision	V				
Project Administration	V	V	V	V	V
Funding Acquisition	V	V	V	V	V

Poster-Related Contributions					
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Poster	V			V	
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PowerPoint	V	V			
Preparation					
Oral	V		V		V
Presentation					